Graphic Era

# MID TERM EVEN SEMESTER EXAMINATION-2026

Roll no. 2594102

**Name of the Program & Semester:** B.Tech II SEM.
**Name of the Course:** Programming in C
**Course Code:** TCS202
**Time:** 1.5 hours

**Maximum Marks:** 50

**Note:**
(i) Answer all the questions except Que(5) by choosing any one of the sub-questions.
(ii) All 5 sub-questions in Que(5) are compulsory.
(iii) Each question carries 10 marks.

| | | |
|---|---|---|
| **Que(1)** | **(10 marks)** | |
| a) | Explain the structure of a C program. Discuss the following terms:<br>1. break and continue<br>2. precedence and associativity<br>3. qualifiers<br>4. typecasting | CO1 |
| | **OR** | |
| b) | Explain the Logical, Increment/Decrement, Bitwise and Ternary operators in C. Write a C program to find the roots of a quadratic equation for the discriminant equal to zero and greater than zero. | |
| **Que(2)** | **(10 marks)** | |
| a) | Draw a flowchart to print following pattern.<br>1<br>0 1<br>1 0 1<br>0 1 0 1<br>1 0 1 0 1 | CO2 |
| | **OR** | |
| b) | Draw a flowchart to read day number from the user and print the next day of the week. Implement a C program for the same using a switch-case statement.<br>Assume: 1-Sunday, 2-Monday, 3-Tuesday, 4-Wednesday, 4-Thursday, 5-Friday, 6-Friday, 7-Saturday.<br>Sample Input: Inputted day number: 3<br>Sample Output: Output next day: Wednesday | |
| **Q3.** | **(10 marks)** | |
| a) | Explain the concept of recursion in C. How it is different from iteration? Discuss at least three points. Develop a recursive function in C to find the sum of following series.<br>**1 - 3 + 5 - 7 + 9 ..... N (where N is odd)** | CO4 |
| | **OR** | |
| b) | Describe the concept of Storage Classes in C. Discuss in detail with syntax and examples, also compare them using Storage Area, Initial value, Scope and Lifetime. | |

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**Q4.** (10 marks)

a) Develop a C code which passes a positive integer number into a function *check()*. If the number is divisible by 3 then the function returns its sum of digit and if not, then the function returns its multiplication of digits. **CO4**

**OR**

b) Develop a C code which passes an integer number into a function *strong()* and checks whether the number is a strong number or not. Create a separate function *fact()* to calculate factorial. Note: Strong number is a number whose sum of factorial of each digit is equivalent to that number. For eg. 145(1! + 4! +5!)

**Q5.** Predict the output of the following code and justify your answer. Assume below code snippets are free from syntax errors. (10 marks) **CO1 CO3**

i)
```
#include<stdio.h>
int main()
{
   int a = 5, b = 3;
   printf("%d ", a+++++b);
   printf("%d %d", a, b);
   return 0;
}
```

ii)
```
#include<stdio.h>
int main()
{
   int x = 10, y = 5, z;
   z = x > y > 2;
   printf("%d", z);
   return 0;
}
```

iii)
```
#include<stdio.h>
void Disp(int x)
{
static int X=5;
  printf("%d %d \n", x, X);
  --X;
}
int main()
{  int i;
   static int x=4;
   for(i=0;i<3;i++)
       Disp(x++);
   return 0;
}
```

iv)
```
#include<stdio.h>
int main()
{
   int n = 2;
   switch(n)
   {
      case 1: printf("One ");
      case 2: printf("Two ");
      case 3: printf("Three ");
      default: printf("Default");
   }
   return 0;
}
```

v)
```
#include<stdio.h>
int Mod(int x, int y)
{
        if(y == 0)
                return x;
        return(Mod(y, x%y));
}
int main()
{
   printf("%d", Mod(10, 35));
   return 0;
}
```